Short Stature, Dauber-Argente Type (PAPP-A2 Deficiency): A Comprehensive Disease Characterization

Disease: Short Stature, Dauber-Argente Type (SSDA) — PAPP-A2 deficiency **OMIM:** #619489 · **MONDO:** MONDO:0859182 · **Gene:** *PAPPA2* (OMIM *619485, HGNC:14615, 1q25.2) **Category:** Mendelian, autosomal recessive · **Inheritance:** Autosomal recessive

Summary

Short Stature, Dauber-Argente type (SSDA) is an ultra-rare autosomal recessive Mendelian growth disorder caused by biallelic loss-of-function mutations in *PAPPA2*, the gene encoding **pappalysin-2 (PAPP-A2)**, a metalloproteinase that cleaves insulin-like growth factor binding proteins IGFBP-3 and IGFBP-5 to release bioactive IGF-1 from its circulating reservoirs. First described by Dauber, Argente and colleagues in 2016, the disorder defines a novel mechanism of growth failure: not IGF-1 *deficiency*, but IGF-1 *sequestration*. Because PAPP-A2 proteolysis is abolished, IGF-1 remains trapped in high-molecular-weight ternary complexes (IGF-1/IGFBP-3 or -5/ALS), producing the paradoxical and diagnostically distinctive biochemical signature of **elevated total IGF-1, IGFBP-3, IGFBP-5, ALS and IGF-II, but reduced free (bioactive) IGF-1**.

Clinically, affected individuals are typically of normal size at birth and then develop **progressive postnatal growth failure**, accompanied by moderate microcephaly, thin long bones, mildly decreased bone mineral density (BMD), and insulin resistance. The condition has been reported in a small number of consanguineous families (Spanish and Saudi kindreds), consistent with private recessive mutations. The founding families carried the homozygous variants p.D643fs25* and p.Ala1033Val, with a third family confirming the syndrome via a homozygous nonsense mutation (p.Glu886*).

Because the fundamental defect is impaired *liberation* of IGF-1 rather than an absolute lack of IGF-1, the disorder is **therapeutically tractable with recombinant human IGF-1 (rhIGF-1, mecasermin)**, which bypasses the proteolytic bottleneck. Long-term (6-year) treatment increased growth velocity and bioactive IGF-1, allowed both treated siblings to reach target

height, normalized BMD, increased lean mass, and improved insulin sensitivity, with no hypoglycemia or other adverse effects observed. Recombinant PAPP-A2 (rhPAPP-A2) itself is an emerging, mechanistically direct experimental therapy, with murine data suggesting sex-specific (female-predominant) efficacy. The *Pappa2* knockout mouse faithfully recapitulates the postnatal growth retardation and reduced bone length of the human disease.

Key Findings

Finding 1 — Biallelic loss-of-function *PAPPA2* mutations cause SSDA

The disease is caused by homozygous (biallelic) loss-of-function mutations in *PAPPA2*. The seminal 2016 report described two unrelated families carrying the homozygous mutations **p.D643fs25*** (a frameshift) and **p.Ala1033Val** (a missense), both associated with progressive postnatal growth failure. Critically, *in vitro* IGFBP cleavage assays demonstrated that **both mutations cause a complete absence of PAPP-A2 proteolytic activity**, establishing loss of enzymatic function as the disease mechanism rather than reduced expression or altered substrate binding alone.

> *"Two different homozygous mutations in PAPPA2, p.D643fs25* and p.Ala1033Val, were associated with this novel syndrome of growth failure. In vitro analysis of IGFBP cleavage demonstrated that both mutations cause a complete absence of PAPP-A2 proteolytic activity."* — PMID: 26902202

A third, independent kindred from Saudi Arabia subsequently confirmed the syndrome and expanded the mutation spectrum with a new homozygous nonsense mutation, **p.Glu886*** in exon 7, in two siblings with postnatal growth retardation and decreased IGF-1 availability.

> *"two siblings of a third family from Saudi Arabia with postnatal growth retardation and decreased IGF1 availability due to a new homozygous nonsense mutation (p.Glu886* in exon 7) in PAPPA2"* — PMID: 34272725

Together these independent families establish autosomal recessive inheritance with a variant spectrum dominated by truncating (frameshift, nonsense) and inactivating missense alleles, all converging on complete loss of proteolytic activity. **Variant classification:** pathogenic (ACMG/AMP), functional consequence = loss of function, germline origin.

Finding 2 — Mechanism: IGF-1 is trapped in ternary complexes, lowering free IGF-1 despite elevated total IGF-1

The pathophysiology is a disorder of IGF-1 bioavailability. PAPP-A2 normally cleaves IGFBP-3 and IGFBP-5, the high-affinity binding proteins that, together with the acid-labile subunit (ALS), form the ~150 kDa ternary complex that serves as the circulating reservoir of IGF-1. When PAPP-A2 activity is lost, IGF-1 cannot be liberated from these complexes. Affected patients therefore show **elevated circulating total IGF-1, IGFBP-3, IGFBP-5, ALS, and IGF-II**, yet **decreased free (bioactive) IGF-1** — the biochemical hallmark of the disease.

> *"Multiple members of two unrelated families presented with progressive growth failure, moderate microcephaly, thin long bones, mildly decreased bone density and elevated circulating total IGF-I, IGFBP-3, and -5, acid labile subunit, and IGF-II concentrations."* — PMID: 26902202

Size-exclusion chromatography provided direct mechanistic proof, showing a significant increase in IGF-1 bound in its ternary complex and correspondingly decreased free IGF-1:

> *"Size-exclusion chromatography showed a significant increase in IGF-I bound in its ternary complex. Free IGF-I concentrations were decreased. These patients provide important insights into the regulation of longitudinal growth in humans, documenting the critical role of PAPP-A2 in releasing IGF-I from its BPs."* — PMID: 26902202

This makes SSDA a canonical example of a "hormone availability" disorder: total hormone measurements are misleadingly high, and only the free/bioactive fraction reflects the true endocrine deficit at the tissue level.

Finding 3 — rhIGF-1 therapy improves growth, height, and bone mineral density

Because the defect is sequestration rather than deficiency, exogenous IGF-1 bypasses the proteolytic bottleneck. Two Spanish siblings (homozygous p.D643fs25*) treated with progressively escalated rhIGF-1 (40–120 μg/kg twice daily) showed a clear increase in growth velocity and height, together with increased bioactive IGF-1 and diminished spontaneous GH secretion (consistent with restored negative feedback), while total IGF-1 and IGFBP-3 remained elevated.

> *"There was a clear increase in growth velocity and height in both siblings. Bioactive IGF-1 was increased, and spontaneous GH secretion was diminished after acute administration of rhIGF-1, whereas serum total IGF-1 and IGFBP-3 levels remained elevated. No episodes of hypoglycemia or any other secondary effects were observed during treatment."* — PMID: 27648969

Long-term (6-year) follow-up confirmed durable benefit: both patients achieved their target height, BMD progressively normalized, and lean mass increased.

> *"Growth velocity continued to increase and both patients achieved their target height. Free IGF-1 concentrations increased notably after rhIGF-1 administration, with serum IGFBP-3, IGFBP-5 and ALS levels also being higher during treatment. BMD was progressively normalized and an increase in lean mass was also noted during treatment."* — PMID: 34358737

NCIT term suggestion: Mecasermin (recombinant human IGF-1) therapy.

Finding 4 — The *Pappa2* knockout mouse recapitulates postnatal growth retardation

The constitutive *Pappa2* knockout (KO) mouse is a strong model of the human disease. KO mice are normal size at birth but develop postnatal growth retardation, with males ~10% and females ~25–30% lower body weight than wild-type littermates, and reduced adult femur and body length — without significant effects on bone mineral density in the mouse.

> *"The most striking phenotype of the PAPP-A2 KO mouse was postnatal growth retardation. Male and female PAPP-A2 KO mice had 10 and 25-30% lower body weight, respectively, than WT littermates. Adult femur and body length were also reduced in PAPP-A2 KO mice, but without significant effects on bone mineral density."* — PMID: 21586553

PAPP-A2 is highly expressed in placenta, with abundant fetal, skeletal, and reproductive tissue expression, consistent with its role in longitudinal growth. The sex-dimorphic severity in the mouse (females more affected) foreshadows the sex-specific therapeutic responses discussed below.

Finding 5 — SSDA causes insulin resistance and low bone mineral density, both rhIGF-1-responsive

Beyond short stature, patients demonstrate **insulin resistance** and **below-average bone mineral density**, both of which improve on rhIGF-1.

> *"Additionally, the patients demonstrated insulin resistance and below average bone mineral density (BMD). The PAPP-A2 deficient patients were treated with recombinant human IGF-1, resulting in improvements in growth velocity, insulin resistance, and BMD."* — PMID: 29280739

Untargeted GC-MS metabolomics of the treated siblings revealed that rhIGF-1 most strongly altered **free fatty acid and amino acid pathways**, implicating lipid and protein metabolism as the primary systemic metabolic targets of restored IGF-1 signaling.

> *"Free fatty acids (FFAs) and amino acids showed the largest changes in the compared metabolic profiles, suggesting that rhIGF1 treatment has the greatest effects on lipid and protein metabolic pathways in the PAPP-A2 deficient subjects."* — PMID: 30119035

Finding 6 — PAPP-A2 acts within the IGFBP–STC2–PAPP-A (ISPa) axis; rhPAPP-A2 is an emerging targeted therapy

PAPP-A2 is one node of a broader regulatory network governing IGF bioavailability. It cleaves IGFBP-3 and -5, and its activity is inhibited by **stanniocalcin-2 (STC2)**. The related protease PAPP-A cleaves IGFBP-4 and -5 and is likewise STC2-inhibited, together forming the **IGFBP4–STC2–PAPP-A (ISPa) axis**.

> *"PAPP-A2 is a protease which cleaves IGFBP-3 and -5, while STC2 inhibits PAPP-A and PAPP-A2 activity."* — PMID: 29280739

In *Pappa2*-deficient mice, recombinant rhPAPP-A2 (compared alongside rhGH and rhIGF-1) modulated growth-related IGF-1 signaling with sex-specific effects, supporting rhPAPP-A2 as a mechanistically direct, emerging therapeutic — with efficacy that appears female-predominant.

> *"pointing to rhPAPP-A2 as a promising drug to alleviate postnatal growth retardation underlying low IGF1 bioavailability in a female-specific manner"* — PMID: 38589872

Finding 7 — Diagnostic biochemical signature and the value of circulating PAPP-A2 measurement

Across all reported families, the diagnostic pattern is consistent: **elevated total IGF-1, IGFBP-3, IGFBP-5 (variable), and ALS with decreased free IGF-1**, plus progressive postnatal short stature and moderate microcephaly. Measuring circulating PAPP-A2 directly can focus the workup, because affected individuals have very low or undetectable levels.

> *"pediatric endocrinologists should measure circulating PAPP-A2 levels in the study of short stature as very low or undetectable levels of this protein can help to focus the diagnosis and treatment"* — PMID: 34272725
>
> *"high circulating levels of total IGF1, IGFBP3, and the IGF acid-labile subunit (IGFALS), with decreased free IGF1 concentrations"* — PMID: 34272725

Definitive confirmation is molecular (WES or targeted *PAPPA2* sequencing) demonstrating biallelic mutations, ideally supported by demonstration of absent *in vitro* proteolytic activity.

Finding 8 — Differential diagnosis from other IGF ternary-complex disorders

SSDA is distinguished from the closest mimic, **ALS (IGFALS) deficiency**, by its IGF-1 profile. ALS deficiency shows **low** IGF-1 and disproportionately low IGFBP-3 with mild growth retardation, normal GH-stimulation response, a failed IGF generation test, and insulin insensitivity. PAPP-A2 deficiency shows the opposite IGF-1 direction: **high** total IGF-1/IGFBP-3/ALS but **low** free IGF-1. Both are autosomal recessive and share mild-to-moderate postnatal growth failure and insulin resistance.

> *"complete ALS deficiency is characterized by severe reduction of IGF-I and IGFBP-3 that remain low after GH treatment, associated with mild growth retardation, much less pronounced than the IGF-I deficit. Pubertal delay in boys and insulin insensitivity are common findings"* — PMID: 20679994

Finding 9 — Verified disease and gene identifiers

Identifiers were verified against OMIM, EBI OLS4 (MONDO), HGNC REST, and NIH GTR.

Entity	Identifier
Disease (OMIM)	#619489 — SHORT STATURE, DAUBER-ARGENTE TYPE (SSDA)
Disease (MONDO)	MONDO:0859182
Disease (UMLS/GTR)	C5561968
Gene (OMIM)	*619485 (*PAPPA2*)
Gene (HGNC)	HGNC:14615
Gene (NCBI)	60676
Gene (Ensembl)	ENSG00000116183
Protein (UniProt)	Q9BXP8 (pappalysin-2)
Cytogenetic location	1q25.2

No dedicated Orphanet (ORDO), ICD-10/ICD-11, or MeSH term exists specifically for SSDA; it maps generically to growth-disorder categories (e.g., ICD-10 E34.3).

Mechanistic Model / Interpretation

The disease can be understood as a single-point failure in the IGF-1 liberation cascade. The causal chain runs from genotype through impaired proteolysis to reduced tissue-level IGF-1 receptor signaling and, finally, to reduced longitudinal bone growth.

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Biallelic LOF PAPPA2 mutation (e.g., p.D643fs25*, p.Ala1033Val, p.Glu886*)
                        │
                        ▼
       Complete loss of PAPP-A2 proteolytic activity
                        │
                        ▼
 IGFBP-3 and IGFBP-5 are NOT cleaved (uncleaved BPs accumulate)
                        │
                        ▼
 IGF-1 remains trapped in the 150 kDa ternary complex
 (IGF-1 / IGFBP-3 or -5 / ALS)  ── reservoir cannot be "opened"
                        │
          ┌─────────────┴─────────────┐
          ▼                           ▼
↑ Total IGF-1, IGFBP-3,              ↓ FREE (bioactive) IGF-1
  IGFBP-5, ALS, IGF-II                      │
(paradoxical labs)                          ▼
                            ↓ IGF-1 receptor signaling at growth plate,
                              bone, and metabolic tissues
                                            │
                                            ▼
   Progressive postnatal growth failure · microcephaly · thin long
   bones · low BMD · insulin resistance
```

Upstream vs downstream. The upstream lesion is the loss of proteolytic activity; the elevated total binding proteins and IGF-1 are a *proximal biochemical consequence* (the reservoir backs up), while the low free IGF-1 is the *functionally decisive downstream* event that produces the clinical phenotype. This is why total IGF-1 is a poor guide to disease state and free IGF-1 (or bioactive IGF-1 assays) is essential.

Regulatory context (ISPa axis). PAPP-A2 operates in parallel with PAPP-A, both under negative control by STC2. This explains why the axis is physiologically tuned (e.g., PAPP-A/PAPP-A2 levels shift with obesity, exercise, meals, and other growth states) and why perturbations elsewhere in the axis (obesity, Prader-Willi syndrome, GH deficiency) produce related but distinct IGF-bioavailability phenotypes.

Therapeutic logic. Two rational strategies follow directly from the mechanism: 1. **Bypass** the block by delivering exogenous free IGF-1 (rhIGF-1 / mecasermin) — clinically validated, safe, and effective. 2. **Replace** the missing enzyme with rhPAPP-A2 to restore physiological IGF-1 liberation — mechanistically direct, currently experimental (mouse-stage), with apparent female-predominant efficacy.

Comparative table — IGF ternary-complex short-stature disorders:

Feature	PAPP-A2 deficiency (SSDA)	ALS (IGFALS) deficiency
Gene	*PAPPA2* (1q25.2)	*IGFALS*
Inheritance	Autosomal recessive	Autosomal recessive
Total IGF-1	**High**	**Low**
IGFBP-3	High	Low / undetectable
ALS	High	Low / undetectable
Free IGF-1	Low	Low
Growth failure	Progressive postnatal, moderate	Mild
Insulin	Insulin resistance	Insulin insensitivity
Response to rhIGF-1	Growth, BMD, insulin sensitivity improve	—

Evidence Base

PMID	Title (abbrev.)	Role in this report
26902202	*Mutations in PAPP-A2 cause short stature due to low IGF-I availability*	Founding paper; identifies the two founding mutations, proves complete loss of proteolytic activity, documents the ternary-complex mechanism (F1, F2)
34272725	*PAPP-A2 deficiency in a Saudi family*	Third independent family (p.Glu886*); establishes circulating PAPP-A2 as a diagnostic biomarker (F1, F7)
27648969	*rhIGF-1 improves growth in PAPP-A2 deficiency*	Initial rhIGF-1 treatment response with increased bioactive IGF-1, no adverse effects (F3)
34358737	*Adult height and long-term outcomes after rhIGF-1*	6-year outcome: target height reached, BMD normalized (F3)
21586553	*PAPP-A2 KO mouse*	Model organism recapitulating postnatal growth retardation, reduced bone length (F4)
29280739	*Novel modulators PAPP-A2 and STC2*	Insulin resistance/low BMD and their rhIGF-1 response; defines STC2 inhibition (F5, F6)
30119035	*Metabolomics of rhIGF1 treatment*	Lipid/protein metabolic pathway changes with therapy (F5)
38589872	*Sex-based differences: rhGH, rhIGF1, rhPAPP-A2*	Emerging rhPAPP-A2 therapy, female-specific efficacy (F6)
20679994	*ALS deficiency*	Differential diagnosis anchor (F8)
33919940	*rmIGF-1 sex-specific bone effects in Pappa2 mice*	Supports sex-specific bone remodeling responses (context for F4/F6)
41528724	*IGF-I bioavailability in isolated GHD (ISPa)*	Contextualizes the ISPa regulatory axis in a distinct disorder
38662803	*Pappalysins in childhood obesity*	Physiological regulation of the axis (contrast state)
28964325	*Regulation of the IGFBP-4/ STC-2/PAPP-A axis*	Physiological modulators of the axis
38141219	*Pappalysins in Prader-Willi syndrome*	Axis behavior in a related growth disorder

PMID	Title (abbrev.)	Role in this report
28445628	*ACLSD Latin American families*	Expands ALS deficiency phenotype for differential diagnosis
30717585	*Novel homozygous ALS mutation*	Additional ALS deficiency reference

Evidence source types: Human clinical (case series/family studies: 26902202, 34272725, 27648969, 34358737, 29280739, 30119035); model organism (21586553, 38589872, 33919940); *in vitro* (IGFBP cleavage assays within 26902202); physiological/observational human cohorts (38662803, 28964325, 38141219, 41528724).

Section-by-Section Data Compilation

1. Disease Information

A concise overview: SSDA is an ultra-rare Mendelian, autosomal recessive endocrine growth disorder of impaired IGF-1 bioavailability. **Synonyms:** PAPP-A2 deficiency; pappalysin-2 deficiency; short stature due to PAPP-A2 deficiency; SSDA. **Identifiers:** OMIM #619489; MONDO:0859182; UMLS C5561968; no dedicated Orphanet/ICD/MeSH term. Information is derived from aggregated disease-level resources plus individual patient family case series (not EHR-scale).

2. Etiology

Causal factor: genetic — biallelic loss-of-function *PAPPA2* mutations. **Genetic risk factors:** consanguinity (all reported families are consanguineous, yielding homozygous private mutations); carriers are heterozygous and unaffected. **Environmental / infectious / lifestyle factors:** none established as causal. **Protective factors:** none characterized. **Gene–environment interactions:** none established; the ISPa axis is physiologically modulated by nutrition, obesity, exercise and meals (P 28964325), P 38662803 which could theoretically modify expressivity but is unproven in SSDA.

3. Phenotypes (with HPO suggestions)

Phenotype	Type	HPO term	Onset / severity / frequency
Short stature / postnatal growth failure	Clinical sign	HP:0004322 (Short stature); HP:0008897 (Postnatal growth retardation)	Postnatal, progressive; moderate–severe; ~all patients
Microcephaly	Physical	HP:0000252 (Microcephaly)	Congenital/early; moderate; reported in most
Thin long bones / gracile bones	Radiographic sign	HP:0003100 (Slender long bone)	Childhood; mild–moderate
Decreased bone mineral density	Lab/imaging	HP:0004349 (Reduced bone mineral density)	Childhood; mild; rhIGF-1-responsive
Insulin resistance	Lab abnormality	HP:0000855 (Insulin resistance)	Childhood; variable; rhIGF-1-responsive
Elevated circulating IGF-1	Lab abnormality	HP:0030269 (Increased circulating IGF-1)	Persistent hallmark

Quality of life impact: primarily driven by short stature (psychosocial), and skeletal/metabolic morbidity; substantially improvable with rhIGF-1.

4. Genetic / Molecular Information

Causal gene: *PAPPA2* (HGNC:14615; OMIM *619485; 1q25.2; UniProt Q9BXP8). **Pathogenic variants:** p.D643fs25* (frameshift), p.Ala1033Val (missense), p.Glu886* (nonsense) — all homozygous, all pathogenic, all complete loss of function. **Allele frequency:** private/ultra-rare, essentially absent from population databases (gnomAD). **Origin:** germline. **Functional consequence:** loss of function (abolished proteolytic activity). **Modifier genes:** none confirmed; STC2 is a physiological inhibitor of PAPP-A2 activity and a candidate modulator. **Epigenetic / chromosomal abnormalities:** none reported.

5. Environmental Information

No environmental, lifestyle, or infectious contributors. The disease is fully genetic.

6. Mechanism / Pathophysiology

Molecular pathway: GH–IGF-1 axis; IGFBP proteolysis (metalloprotease/pappalysin activity). **Cellular processes:** longitudinal bone growth at the growth plate (chondrocyte proliferation), bone remodeling, insulin/glucose metabolism. **Protein dysfunction:** loss of metalloproteinase function of pappalysin-2. **Biochemical abnormality:** failure to cleave IGFBP-3/IGFBP-5, trapping IGF-1 in ternary complexes. **Metabolic changes:** rhIGF-1 predominantly affects free fatty acid and amino acid metabolism (P 30119035).

GO term suggestions: GO:0008233 (peptidase activity), GO:0004222 (metalloendopeptidase activity), GO:0043568 (positive regulation of insulin-like growth factor receptor signaling pathway), GO:0060348 (bone development), GO:0030282 (bone mineralization). **CHEBI:** IGF-1 (peptide hormone). **CL terms:** CL:0000138 (chondrocyte), CL:0000062 (osteoblast).

7. Anatomical Structures Affected

Primary: skeletal system — long bones and growth plates (UBERON:0002481 bone tissue; UBERON:0006255 epiphyseal plate), skull/head (microcephaly, UBERON:0000033 head). **Secondary/systemic:** endocrine (GH–IGF axis), metabolic tissues (insulin resistance). **Subcellular:** secreted/extracellular protease acting in blood plasma (GO:0005576 extracellular region). **Lateralization:** bilateral/symmetric (systemic endocrine disorder).

8. Temporal Development

Onset: normal size at birth; **postnatal**, progressive growth failure emerging in infancy/childhood (insidious, chronic). **Progression:** slowly progressive short stature over childhood; **chronic lifelong** biochemical defect. **Critical period / window of opportunity:** childhood and pre-pubertal growth window — rhIGF-1 initiated during active growth allowed attainment of target adult height (P 34358737).

9. Inheritance and Population

Inheritance: autosomal recessive. **Penetrance:** appears complete in biallelic homozygotes; heterozygous carriers unaffected. **Expressivity:** variable severity, with possible sex differences (mouse data show females more affected). **Consanguinity:** central — all reported families consanguineous. **Founder effects:** private family-specific mutations rather than shared founders. **Epidemiology:** ultra-rare; only a handful of families reported worldwide (Spanish, Saudi); precise prevalence/incidence not established. **Carrier frequency:** not established (essentially absent from population databases).

10. Diagnostics

Biochemical signature (key screen): high total IGF-1, IGFBP-3, IGFBP-5, ALS, IGF-II; **low free/bioactive IGF-1**. **Direct biomarker:** low/undetectable circulating PAPP-A2 (P 34272725).

Confirmatory genetic testing: WES or targeted *PAPPA2* single-gene/panel sequencing demonstrating biallelic variants; functional confirmation via *in vitro* IGFBP cleavage assay (absent proteolysis). **Imaging:** radiographs (thin long bones), DXA (low BMD). **Differential diagnosis:** ALS (IGFALS) deficiency (low IGF-1), GH deficiency/insensitivity, IGF-1/IGF1R defects, other IGF-axis short-stature syndromes. **LOINC:** IGF-1, IGFBP-3, free IGF-1 assays.

11. Outcome / Prognosis

No excess mortality reported; the disorder is not life-threatening. **Morbidity:** short stature, low BMD, insulin resistance; psychosocial impact of short stature. **Recovery potential:** strongly favorable with rhIGF-1 — target height achieved, BMD normalized, insulin sensitivity improved, lean mass increased over 6-year treatment (P 34358737 P 29280739).

Prognostic factors: early initiation of rhIGF-1 during the growth window; free (not total) IGF-1 as the treatment-monitoring biomarker.

12. Treatment

Established: recombinant human IGF-1 (mecasermin), 40–120 μg/kg twice daily, escalated — improves growth velocity, height, BMD, insulin resistance; safe (no hypoglycemia observed) (P 27648969 P 34358737 P 29280739).

Emerging/experimental: recombinant PAPP-A2 (rhPAPP-A2) — mechanistically direct enzyme

replacement; mouse-stage; female-predominant efficacy (P 38589872).
rhGH is comparatively ineffective because the bottleneck is downstream of GH. **NCIT suggestions:** Mecasermin; recombinant IGF-1 therapy. **Supportive:** monitoring of growth, BMD, glucose/insulin; genetic counseling.

13. Prevention

No primary prevention (genetic disorder). **Secondary prevention:** early biochemical screening in short-stature workups (measure free IGF-1 and circulating PAPP-A2 when total IGF-1 is paradoxically high). **Genetic prevention/counseling:** carrier and cascade testing in consanguineous families; prenatal/preimplantation testing feasible once the familial variant is known. **Tertiary prevention:** rhIGF-1 to forestall growth and skeletal complications.

14. Other Species / Natural Disease

Taxonomy: *Mus musculus* (NCBI:txid10090) used as model. **Orthologous gene:** murine *Pappa2*. No naturally occurring companion-animal or wildlife disease has been characterized (no OMIA entry established here). Disease mechanism (IGF-1 liberation via pappalysins) is evolutionarily conserved.

15. Model Organisms

Primary model: constitutive *Pappa2* knockout mouse (P 21586553)
— mammalian, genetic knockout. **Phenotype recapitulation:** strong — normal birth size then postnatal growth retardation, reduced femur and body length; sex-dimorphic severity (females more affected). **Limitations:** mouse KO shows no significant BMD reduction (unlike human low BMD); conditional/tissue-specific models not central to reported work. **Applications:** dissecting IGF-1 bioavailability; testing rhIGF-1/rhGH/rhPAPP-A2 (P 38589872 P 33919940).
Resources: MGI.

Limitations and Knowledge Gaps

- **Extremely small patient base.** Conclusions rest on a handful of consanguineous families (Spanish, Saudi). Prevalence, incidence, carrier frequency, penetrance, and the full phenotypic spectrum are not statistically established.
- **No formal Orphanet/ICD/MeSH coding**, complicating registry-based epidemiology and case-finding.
- **Treatment evidence is from uncontrolled case series**, not randomized trials. Long-term (adult) metabolic and skeletal outcomes beyond 6 years are unknown.
- **Sex-specific effects** are strongly suggested by mouse data but not systematically quantified in humans.
- **rhPAPP-A2 remains pre-clinical**; human safety, immunogenicity, dosing, and efficacy are untested.
- **Modifier genetics and the role of the STC2/ISPa axis** in modulating human disease severity are unexplored.
- **BMD discrepancy** between human (low BMD) and mouse KO (BMD unaffected) indicates the mouse does not fully capture the human skeletal phenotype.

Proposed Follow-up Experiments / Actions

1. **Establish an international patient registry** for *PAPPA2* biallelic cases to define natural history, prevalence, and genotype–phenotype correlations, and to enable prospective outcome tracking.
2. **Prospective, standardized rhIGF-1 dose-response and monitoring study** using free/bioactive IGF-1 as the primary pharmacodynamic marker; assess adult height, BMD, insulin sensitivity, and lean mass with pre-specified endpoints.
3. **Advance rhPAPP-A2 toward clinical translation:** IND-enabling pharmacokinetics/pharmacodynamics, immunogenicity, and sex-stratified efficacy studies in *Pappa2* KO mice, then first-in-human evaluation.
4. **Deploy circulating PAPP-A2 assays** as a first-line screen in pediatric short-stature clinics where total IGF-1 is paradoxically elevated with low free IGF-1, and validate assay cut-offs.
5. **Investigate ISPa-axis modifiers (STC2, PAPP-A, IGFBP-4)** as potential disease modulators and alternative druggable nodes.

6. **Develop humanized or conditional mouse models** and iPSC-derived chondrocyte/osteoblast systems to resolve the BMD discrepancy and study cell-type-specific IGF-1 signaling at the growth plate.

Consensus Answer

Short Stature, Dauber-Argente type (OMIM #619489; MONDO:0859182; PAPP-A2 deficiency) is an ultra-rare autosomal recessive disorder caused by biallelic loss-of-function mutations in *PAPPA2* (1q25.2), encoding the metalloproteinase pappalysin-2 that cleaves IGFBP-3 and IGFBP-5 to release bioactive IGF-1. Loss of this proteolysis traps IGF-1 in circulating ternary complexes, producing the hallmark of high total IGF-1/IGFBP-3/ALS but low free IGF-1, and causing progressive postnatal growth failure with microcephaly, thin long bones, low bone mineral density, and insulin resistance. Because the defect is IGF-1 sequestration rather than deficiency, recombinant human IGF-1 (mecasermin) safely restores growth, bone density, and insulin sensitivity, with recombinant PAPP-A2 an emerging experimental therapy.

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